3.4 Materials

This section examines the physical properties of springs and materials.

Learners can carry out a range of experimental work to enhance their knowledge and skills, including the management of risks and analysis of data to provide evidence for relationships between physical quantities. There are opportunities to consider the selection of appropriate materials for practical applications (HSW5, 6, 8, 9, 12).

3.4.1 Springs

		Learning outcomes	Additional guidance
		Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)		tensile and compressive deformation; extension and compression	
(b)		Hooke's law	
(c)		force constant k of a spring or wire; $F = kx$	
(d)	**(i)**	force–extension (or compression) graphs for springs and wires	*M3.2*
	(ii)	techniques and procedures used to investigate force–extension characteristics for arrangements which may include springs, rubber bands, polythene strips.	**PAG2** HSW5, 6